RKHS of $K(x, y) = (c + x^\top y)^3$

1. Expansion via Binomial Theorem

Applying the binomial theorem with $p = 3$:

$$K(x, y) = (c + x^\top y)^3 = c^3 + 3c^2(x^\top y) + 3c(x^\top y)^2 + (x^\top y)^3.$$

Each power of $x^\top y$ lifts to a tensor inner product. Writing $\langle A, B\rangle_F = \mathrm{tr}(A^\top B)$ for the Frobenius inner product:

$$(x^\top y)^k = \langle x^{\otimes k},\, y^{\otimes k}\rangle,$$

where $x^{\otimes k} = x \otimes \cdots \otimes x$ (k times) is a rank-1 symmetric tensor of order k.

2. Feature Map

The explicit feature map $\phi : \mathbb{R}^d \to \mathcal{H}$ is

$$\phi(x) = \begin{pmatrix} c^{3/2} \\ \sqrt{3}\, c\, x \\ \sqrt{3c}\, x^{\otimes 2} \\ x^{\otimes 3} \end{pmatrix},$$

where $x^{\otimes 2} = xx^\top \in \mathbb{R}^{d\times d}$ and $x^{\otimes 3} \in \mathbb{R}^{d\times d\times d}$, both flattened. Verification:

$$\langle \phi(x), \phi(y)\rangle = c^3 + 3c^2(x^\top y) + 3c(x^\top y)^2 + (x^\top y)^3 = K(x, y). \qquad \checkmark$$

3. The RKHS as a Tensor Space

Every $f \in \mathcal{H}_K$ is parametrised by $(a,\, b,\, C,\, \mathcal{T})$ with

$$a \in \mathbb{R}, \quad b \in \mathbb{R}^d, \quad C \in \mathbb{R}^{d\times d}, \quad \mathcal{T} \in \mathbb{R}^{d\times d\times d},$$

and evaluated as

$$\boxed{f(x) = a + b^\top x + \mathrm{tr}(C^\top xx^\top) + \langle \mathcal{T},\, x^{\otimes 3}\rangle.}$$

The cubic term satisfies $\langle \mathcal{T},\, x^{\otimes 3}\rangle = x^\top \mathcal{T}_{(1)}(x \otimes x)$, where $\mathcal{T}_{(1)} \in \mathbb{R}^{d\times d^2}$ is the mode-1 matricisation of $\mathcal{T}$.

4. RKHS Inner Product

For $f \leftrightarrow (a, b, C, \mathcal{T})$ and $g \leftrightarrow (a', b', C', \mathcal{T}')$:

$$\boxed{\langle f,\, g\rangle_{\mathcal{H}} = \frac{aa'}{c^3} + \frac{b^\top b'}{3c^2} + \frac{\langle C,\, C'\rangle_F}{3c} + \langle \mathcal{T},\, \mathcal{T}'\rangle_F,}$$

where $\langle \mathcal{T}, \mathcal{T}'\rangle_F = \mathrm{tr}(\mathcal{T}_{(1)}^\top \mathcal{T}'_{(1)})$. The RKHS norm is

$$\|f\|_{\mathcal{H}}^2 = \frac{a^2}{c^3} + \frac{\|b\|^2}{3c^2} + \frac{\|C\|_F^2}{3c} + \|\mathcal{T}\|_F^2.$$

1 5. Representer in Tensor Form

The kernel section $K(\cdot, x)$ corresponds to

$$K(\cdot, x) \;\leftrightarrow\; \big(c^3, \quad 3c^2 x, \quad 3c\, xx^\top, \quad x^{\otimes 3}\big).$$

The reproducing property:

$$\begin{aligned}\langle f,\, K(\cdot, x)\rangle_{\mathcal{H}} &= \frac{a \cdot c^3}{c^3} + \frac{b^\top (3c^2 x)}{3c^2} + \frac{\langle C,\, 3c\, xx^\top\rangle_F}{3c} + \langle \mathcal{T},\, x^{\otimes 3}\rangle_F \\ &= a + b^\top x + x^\top C x + \langle \mathcal{T},\, x^{\otimes 3}\rangle \;=\; f(x). \qquad \checkmark\end{aligned}$$

6. Kernel Matrix (No Loops)

On a dataset $X = [x_1 \mid \cdots \mid x_n] \in \mathbb{R}^{d\times n}$, let $\mathbf{G} = c\,\mathbf{1}\mathbf{1}^\top + X^\top X$. The full kernel matrix is the Hadamard cube:

$$\boxed{\mathbf{K} = \mathbf{G} \circ \mathbf{G} \circ \mathbf{G}.}$$

Assembled from one matrix multiply and two elementwise products—no loops required.

7. General Degree-p Pattern

For $K(x, y) = (c + x^\top y)^p$:

$$K(x, y) = \sum_{k=0}^{p} \binom{p}{k} c^{p-k} \langle x^{\otimes k},\, y^{\otimes k}\rangle.$$

Degree k	Feature	Weight in $\|f\|_{\mathcal{H}}^2$	Parameter
0	1	c^{-p}	$a \in \mathbb{R}$
1	x	$\left[\binom{p}{1} c^{p-1}\right]^{-1}$	$b \in \mathbb{R}^d$
2	$x^{\otimes 2}$	$\left[\binom{p}{2} c^{p-2}\right]^{-1}$	$C \in \mathbb{R}^{d\times d}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$
k	$x^{\otimes k}$	$\left[\binom{p}{k} c^{p-k}\right]^{-1}$	$\mathcal{T}^{(k)} \in \mathbb{R}^{d^k}$
p	$x^{\otimes p}$	1	$\mathcal{T}^{(p)} \in \mathbb{R}^{d^p}$

The RKHS is the space of **polynomials of degree** $\leq p$ on $\mathbb{R}^d$, with geometry determined by symmetric tensor algebra and Frobenius inner products—no index gymnastics needed.